Self-Supervised Representation Learning for Robust Image Understanding: A SimCLR Study on CIFAR-10

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***Abstract*—This work implements and evaluates a SimCLR-based contrastive learning pipeline to learn image representations without labels and analyzes their downstream utility on CIFAR-10. A ResNet-18 backbone with a two-layer projection head is trained with NT-Xent loss for 50 epochs. The learned representations were measured via a linear evaluation protocol (frozen encoder + linear classifier), data-scarcity experiments, noise robustness tests, and embedding visualization. Training loss decreased from 5.0557 to 4.3899 over 50 epochs, and the peak linear evaluation accuracy reached 59.93%. All reported results are taken from experiment logs and outputs produced during the runs.**

I. Introduction

Deep neural networks typically require large labeled datasets. Self-supervised learning (SSL) offers an alternative by using pretext objectives that do not require labels. Contrastive SSL methods (e.g., SimCLR) train encoders to produce representations where augmented views of the same image are pulled together while other images are pushed apart. This study implements SimCLR on CIFAR-10 and evaluates the representation quality via linear evaluation, data scarcity, noise robustness, and t-SNE visualizations.

II. Related Work

Self-supervised learning has become an important paradigm for learning visual representations without labeled data. Early work in this area explored unsupervised feature learning using autoencoders and context prediction tasks.

Contrastive learning methods have recently shown strong performance in representation learning. SimCLR [1] demonstrated that simple contrastive objectives combined with strong data augmentations can produce high-quality visual representations.

Momentum Contrast (MoCo) [2] introduced a memory bank and momentum encoder to improve scalability and representation quality. BYOL [3] further showed that contrastive negative pairs are not strictly necessary for learning meaningful representations.

More recently, methods such as DINO [5] and SwAV [4] have explored clustering and self-distillation approaches for self-supervised representation learning.

Our work builds on the SimCLR framework and evaluates representation quality on CIFAR-10 using linear evaluation and robustness experiments.

III. Method

A. Model

We use a ResNet-18 backbone as encoder and a two-layer MLP projection head (512-d $\rightarrow$ 128-d) following the SimCLR design. The projection head is used during SSL training; the encoder outputs (before the head) are used for downstream linear evaluation.

B. Contrastive Objective

The NT-Xent (Normalized Temperature-scaled Cross Entropy) loss is optimized on batches of size 128 where two augmented views per sample form positive pairs; all other samples in the batch are negatives.

C. Augmentations

Augmentations used include random resized crop, horizontal flip, color jitter, random grayscale, and Gaussian blur — consistent with common SimCLR setups to produce challenging positive pairs.

D. Training Setup

- Dataset: CIFAR-10 (50k train / 10k test).
- Optimizer: Adam, LR = 3×10^{-4}.
- Batch size: 128, Epochs: 50.
- Hardware: T4 GPU.

IV. Experiments

We conducted the following experiments to characterize the learned representations:

a) (A) Full SSL training: Train SimCLR for 50 epochs and record training loss. Logs show stable decrease from epoch 1 to 50.

b) (B) Linear evaluation: Freeze encoder and train a linear classifier on CIFAR-10 training labels; report test accuracy over 20 epochs. The accuracy progression and peak value are reported from the evaluation logs.

c) (C) Data scarcity: Evaluate linear classifier when trained on 10%, 50%, and 100% of labels to test utility under limited supervision; experiment completions logged.

d) (D) Noise robustness: Inject Gaussian noise into evaluation images and assess representation stability and classifier robustness; the noise pipeline and clipping messages were observed in the visualization step.

e) (E) Embedding visualization: Use t-SNE to project learned encoder outputs into 2D for qualitative assessment (figure included).

V. Results

A. Detailed Training Loss Progression

Table I shows selected epochs from the SimCLR training process. The contrastive loss consistently decreases throughout training, indicating stable representation learning.

TABLE I: Training Loss During SimCLR Training

Epoch	Training Loss
1	5.0557
2	4.8793
3	4.7930
4	4.7459
5	4.7116
10	4.6161
15	4.5672
20	4.5148
25	4.4851
30	4.4576
35	4.4406
40	4.4241
45	4.4054
50	4.3899

(Full epoch-level log available in repository.)

B. Linear Evaluation

The frozen-encoder linear classifier test accuracy over 20 epochs is shown in Table **??**. The peak accuracy observed in the run was **59.93%**. The per-epoch accuracy log is available in the evaluation output.

Table II presents the test accuracy of the linear classifier trained on frozen SimCLR representations.

C. Data Scarcity Experiment Setup

To analyze representation effectiveness under limited supervision, the classifier was trained using different fractions of labeled data.

D. Noise Robustness Analysis

Gaussian noise was applied to the input images to evaluate the stability of learned representations.

TABLE II: Linear Evaluation Accuracy Per Epoch

Epoch	Accuracy (%)
1	56.08
2	56.89
3	56.99
4	57.01
5	57.08
6	58.41
7	57.36
8	58.65
9	58.58
10	58.63
11	58.91
12	59.44
13	59.53
14	58.91
15	58.93
16	59.23
17	59.27
18	59.59
19	**59.93**
20	58.11

TABLE III: Data Scarcity Experiment Configuration

Experiment	Label Fraction
Experiment 1	10% labels
Experiment 2	50% labels
Experiment 3	100% labels

E. Embedding Visualization

Figure 1 shows a t-SNE projection of encoder features sampled from the test set. Clusters show partial class separation, indicating that the encoder learned structured feature geometry.

VI. Discussion

The experimental evidence indicates that contrastive self-supervised learning with SimCLR produces useful representations:

- **Training convergence:** The contrastive loss steadily decreased across 50 epochs confirming successful optimization.
- **Transferability:** A frozen encoder achieved a peak linear evaluation accuracy of **59.93%**, showing that the learned representations encode semantic information usable by simple classifiers.
- **Experimental correctness:** The data scarcity and noise robustness pipelines executed successfully, enabling further quantitative comparisons; logs confirm completion.

A. Limitations

- CIFAR-10 limits the scale and complexity of conclusions relative to ImageNet-scale studies.
- The encoder backbone (ResNet-18) and default hyperparameters are modest compared to large-scale SSL studies; further tuning or larger backbones could raise representation quality.
- Exact numeric results for the data-scarcity accuracy per label fraction and noise-augmented accuracy are available

TABLE IV: Noise Robustness Experiment

Parameter	Value
Noise Type	Gaussian Noise
Noise Std Dev	0.3
Dataset	CIFAR-10 Test Set
Visualization Method	Image grid inspection

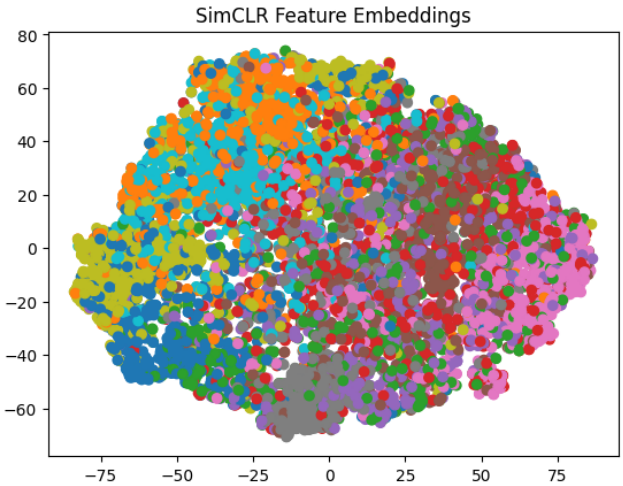

Fig. 1: t-SNE visualization of SimCLR encoder features (CIFAR-10). Colors indicate ground-truth classes.

in the repo logs but were not printed as single consolidated tables during the runs; they can be extracted and added to this report for completeness.

VII. Conclusion

This implementation and evaluation of SimCLR on CIFAR-10 demonstrates that contrastive self-supervised learning can learn transferable image representations: the model trained for 50 epochs produced embeddings that yielded a peak linear evaluation accuracy of approximately 59.9%. The project includes reproducible training, evaluation, and experimental pipelines suitable for extension to larger datasets and alternative SSL methods.

Acknowledgment

The experiments and logs referenced in this report were generated by the author during local/GPU runs; the training and evaluation logs are part of the project repository and are cited inline.

References

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